

Nudging Beyond the Comfort Zone: Efficient Strategy-Guided Exploration for RLVR

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Abstract

Reinforcement learning with verifiable rewards (RLVR) has emerged as a scalable paradigm for improving the reasoning capabilities of large language models. However, its effectiveness is fundamentally limited by exploration: the policy can only improve on trajectories it has already sampled. While increasing the number of rollouts alleviates this issue, such brute-force scaling is computationally expensive, and existing approaches that modify the optimization objective provide limited control over what is explored. In this work, we propose NUDGERL, a framework for structured and diversity-driven exploration in RLVR. Our approach introduces *Strategy Nudging*, which conditions each rollout on lightweight, strategy-level contexts to induce diverse reasoning trajectories without relying on expensive oracle supervision. To effectively learn from such structured exploration, we further propose a unified objective, which decomposes the reward signal into inter- and intra-context components and incorporates a distillation objective to transfer discovered behaviors back to the base policy. Empirically, NUDGERL outperforms standard GRPO with up to $8\times$ larger rollout budgets, while outperforming oracle-guided RL baseline on average across five challenging math benchmarks. These results demonstrate that structured, context-driven exploration can serve as an efficient and scalable alternative to both brute-force rollout scaling and feasibility-oriented methods based on privileged information. Our code is available at <https://github.com/tally0818/NudgeRL>.

1 Introduction

Reinforcement learning with verifiable rewards (RLVR) has emerged as a powerful paradigm for improving the reasoning capabilities of large language models (LLMs) [7, 20]. By leveraging verifiable rewards, methods such as *Group-Relative Policy Optimization* (GRPO) [18] enable scalable post-training without requiring dense supervision. This paradigm has been successfully applied across a wide range of domains.

Despite its success, RLVR remains fundamentally limited by its ability to explore the space of reasoning trajectories. A natural approach is to scale the number of sampled rollouts, which increases the probability of discovering rare trajectories [5]. However, such brute-force scaling quickly becomes computationally prohibitive, motivating alternative approaches that improve exploration efficiency.

Recent work has sought to address this limitation by modifying the optimization objective, for example through entropy regularization or decoupled clipping [24, 26]. While these methods encourage broader exploration at the distribution level, they provide limited control over *what* is explored, and often fail to ensure coverage of semantically meaningful reasoning strategies. Another

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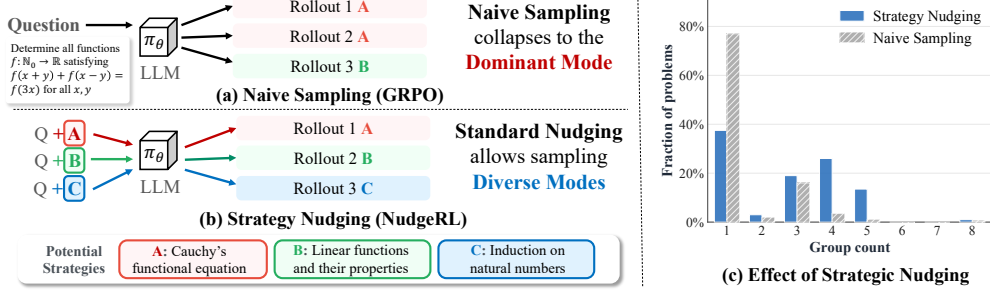


Figure 1: **Concept: Improving exploration diversity through Strategy Nudging.** (a) Naive sampling methods (e.g., GRPO) often collapse to a dominant reasoning mode, limiting the exploration of the reasoning space. (b) NUDGERL introduces Strategy Nudging, which appends lightweight strategy to the input, forcing the model to traverse diverse reasoning modes. (c) As a result, Strategy Nudging significantly increases the number of distinct reasoning approaches discovered compared to the baseline, effectively mitigating the exploration bottleneck. Additional details are in Sec. B

line of work leverages *privileged information*, such as oracle solutions or intermediate reasoning steps, to improve the feasibility of discovering correct trajectories [8, 16, 19, 27]. Although effective, these approaches are primarily feasibility-oriented and rely on strong supervision signals that are expensive to obtain and difficult to scale. Moreover, by guiding the policy toward a narrow set of predefined successful trajectories, they may limit exploration diversity and hinder the discovery of alternative reasoning strategies [23, 25].

In this work, we address the exploration bottleneck by explicitly structuring the reasoning space in a scalable manner. We propose NUDGERL, a framework that introduces *Strategy Nudging* during the exploration phase. Instead of relying on expensive oracle data, Strategy Nudging appends lightweight, heuristic text prompts (e.g., specific strategies for math problems or reasoning keywords) to the input. This deliberately forces the model to traverse distinct, diverse reasoning modes that it might otherwise ignore under purely naive sampling.

However, learning from such context-conditioned exploration introduces new challenges. Since rollouts are generated under different context-conditioned prompts, the samples are naturally partitioned into multiple distinct groups, where reward variation reflects both the intrinsic trajectory quality and context-specific biases, making standard group-wise advantage estimation unreliable. Furthermore, context forcing creates a mismatch between how trajectories are sampled and how the policy is finally used at inference time. Without intervention, improvements discovered under context-forced exploration may not transfer directly to the base policy. To address these challenges, we further introduce (i) an *Inter-Intra group advantage* to enable meaningful credit assignment across context-induced groups, and (ii) a *distillation-augmented objective* that explicitly transfers effective behaviors discovered during context-forced exploration back to the base policy.

Our approach enables structured and diversity-driven exploration while remaining fully compatible with standard RLVR pipelines. Empirically, NUDGERL achieves performance surpassing GRPO even when GRPO is given an $8\times$ larger rollout budget, while outperforming oracle-guided baselines. This demonstrates that scalable, diversity-oriented exploration can serve as an effective alternative to both brute-force rollout scaling and feasibility-driven privileged information.

2 Preliminaries

2.1 Group-Relative Policy Optimization (GRPO)

We consider an empirical distribution of prompts $x_0 \in \mathcal{D}$. For each prompt x_0 , a policy π_θ generates a group of N rollouts $\{y_i\}_{i=1}^N$, where each rollout is sampled as $y_i \sim \pi_\theta(\cdot | x_0)$. Each rollout is evaluated by a verifiable reward function $R(x_0, y_i) \in \{0, 1\}$.

Unlike standard PPO [17], which typically estimates advantages using a learned value function, GRPO [18] derives advantages from group-wise rewards. For rollouts sampled from the same prompt x_0 , let $r_i = R(x_0, y_i)$ denote the reward of rollout y_i . The group-wise advantage is then defined as:

$$\hat{A}_i = \frac{r_i - \mu}{\sigma + \delta}, \quad (1)$$

where μ and σ are the reward mean and standard deviation within the group, and $\delta > 0$ is used for numerical stability. This yields a relative advantage estimate without training a value function.

The policy is then optimized with a PPO-style clipped objective:

$$\mathcal{L}_{\text{GRPO}}(\theta) = -\mathbb{E}_{y \sim \pi_{\theta}(\cdot|x_0)} \left[\min(r_0 \hat{A}, \text{clip}(r_0, 1 - \epsilon, 1 + \epsilon) \hat{A}) \right], \text{ where } r_0 = \frac{\pi_{\theta}(y|x_0)}{\pi_{\text{old}}(y|x_0)}. \quad (2)$$

Thus, GRPO retains PPO’s clipped objective while using group-relative advantages.

2.2 Motivation: From Exploration to Performance Gain

To understand why exploration is a fundamental bottleneck in RLVR, we look beyond trajectory-level rewards and examine how the probability mass of generated tokens shifts during training. Hu et al. [5] characterizes the expected one-step performance improvement (ΔQ_{pos}) in RLVR as:

$$\Delta Q_{\text{pos}} = \frac{\eta}{N} [(1 - S_R) Q_{\text{neg}} A_2 + S_R Q_{\text{pos}} B_2 + S_R (Q_{\text{pos}} U_{\text{neg},2} - Q_{\text{neg}} U_{\text{pos},2})], \quad (3)$$

where Q_{pos} and $Q_{\text{neg}} = 1 - Q_{\text{pos}}$ denote the total probability mass of correct and incorrect tokens, η is the learning rate, and N is the number of rollouts. A_2 and B_2 are the second moments of *sampled* correct and incorrect tokens, while $U_{\text{pos},2}$ and $U_{\text{neg},2}$ are those of *unsampled* correct and incorrect tokens. $S_R \in [0, 1]$ represents the net reward contribution from sampled tokens.

Since $S_R \in [0, 1]$, the first two terms in Eq. 3 are non-negative and drive learning forward. The third term, however, acts as a potential penalty. Because incorrect tokens typically dominate the probability mass ($Q_{\text{neg}} \gg Q_{\text{pos}}$), a large $U_{\text{pos},2}$, meaning the model has significant probability mass on correct trajectories that it simply *fails to explore*, creates a dominant negative force that hinders performance gain. Therefore, the core bottleneck of RLVR lies in the *unexplored* correct regions.

Limitations of rollout scaling. To mitigate this penalty, a naive solution is to increase the rollout size N . Hu et al. [5] shows that for a collection of tokens with probabilities $\{p_i\}$, the expected unsampled second moment after N draws is:

$$\sum_i p_i^2 (1 - p_i)^N, \quad (4)$$

which decreases monotonically with N . However, tokens with small p_i decay slowly, so fully covering long-tail correct trajectories requires prohibitively large rollout budgets.

This highlights the limitation of blindly scaling N to reduce the unexplored correct mass ($U_{\text{pos},2}$). Long-tail correct trajectories remain unlikely to be sampled even under large N , suggesting the need for a *structured exploration* mechanism that can *efficiently* expose such latent trajectories.

3 NUDGERL

We introduce **NUDGERL**, a framework for structured exploration and learning in RLVR. NUDGERL consists of three components: (i) **Strategy Nudging**, which conditions rollout generation on *strategy-level* contexts to induce diverse reasoning trajectories; and (ii) **Inter-intra Group Advantage**, a credit assignment method that enables controlled exploration and exploitation of strategies; and (iii) **Distillation augmented RL objective** to learn from context-conditioned rollouts and distill effective strategies into the policy under the original prompt for inference without external context.

3.1 Strategy Nudging: Structured Exploration via Strategy-Level Contexts

Given that prior work [5] alleviates the exploration bottleneck by reducing unsampled probability mass through larger rollout budgets, a natural question arises: *how many rollouts are required to reliably discover a rare trajectory?* To quantify this discovery cost, consider a rare trajectory y with $\pi(y|x_0) \ll 1$. The expected number of rollouts required to observe y at least once is:

$$\mathbb{E}[\# \text{ rollouts}] = \frac{1}{\pi(y|x_0)}. \quad (5)$$

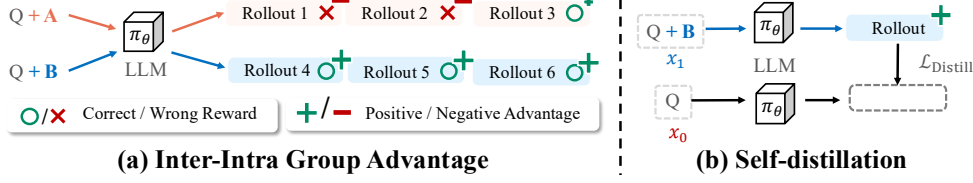


Figure 2: **Overview of the NudgeRL learning mechanism.** (a) **Inter-Intra Group Advantage:** Demonstrates credit assignment that emphasizes reliable contexts (i.e., $\lambda \in (1, 2]$). A successful rollout from a consistently high-reward context (**Strategy B**) receives a larger positive advantage than a rare success from a low-reward context (**Strategy A**). (b) **Self-distillation:** Illustrates bridging the train-test gap. High-quality trajectories discovered via context-conditioned exploration ($Q + B$) are distilled back into the base policy (Q) using $\mathcal{L}_{\text{Distill}}$, allowing the model to internalize effective reasoning modes for context-free inference.

This implies that for low-probability trajectories, the required rollout budget grows prohibitively large. In practice, naive rollout scaling repeatedly samples from high-probability modes of the current policy, leading to diminishing returns in covering rare trajectories.

This motivates conditioning generation on a context c that can shift the sampling distribution toward otherwise rare trajectories. If such a context increases the probability of a trajectory y , i.e., $\pi(y|x, c) \gg \pi(y|x)$, then its expected number of rollouts becomes:

$$\mathbb{E}[\# \text{ rollouts} \mid c] = \frac{1}{\pi(y|x, c)} \ll \frac{1}{\pi(y|x_0)}. \quad (6)$$

Thus, contexts need not provide a solution; they can serve as lightweight controls that alter the sampling distribution and reduce the cost of discovering rare trajectories.

Strategy Nudging. Even though context conditioning can improve exploration efficiency in principle, simply placing multiple contexts in a single prompt leaves the choice of strategy to the policy, which may ignore some contexts and repeatedly follow dominant reasoning patterns. To enforce coverage over contexts, we instead assign a single sampled context to each rollout before generation.

Let $\mathcal{C}(x_0) = \{c_1, \dots, c_M\}$ denote a pool of Strategy-level contexts for the original prompt x_0 . For each rollout index i , we begin with sampling $c^{(i)} \sim \text{Uniform}(\mathcal{C}(x_0))$. To avoid relying exclusively on the context pool and to retain compatibility with the original prompt, we further apply context dropout. Specifically, we sample a mask $b^{(i)} \sim \text{Bernoulli}(1 - p_{\text{drop}})$ and define the context as:

$$z^{(i)} = \begin{cases} c^{(i)}, & b^{(i)} = 1, \\ \emptyset, & b^{(i)} = 0. \end{cases} \quad (7)$$

We then construct the final prompt $x_1^{(i)} = (x_0, z^{(i)})$, and generate $y_i \sim \pi_\theta(\cdot \mid x_1^{(i)})$. By varying $z^{(i)}$ across rollout indices, Strategy Nudging induces diversity at the input-conditioning level, rather than relying solely on sampling from a single prompt. Details on generating $\mathcal{C}$ are in [Sec. B](#).

Context-induced rollout diversity. To verify that Strategy Nudging induces the intended diversity, we compare it against naive sampling without context conditioning. For each prompt, both methods generate 8 rollouts in total: Strategy Nudging samples 4 rollouts from each of 2 contexts without context dropout, whereas the baseline samples all 8 rollouts from the base policy under the original prompt. We then cluster the reasoning structures using an LLM-as-a-judge (gpt-4o-mini [15]) and measure the number of distinct clusters; additional details are provided in [Sec. B](#).

As shown in [Fig. 1](#), Strategy Nudging more often increases the number of distinct reasoning structures relative to naive sampling, whereas the base policy frequently collapses to similar patterns. This suggests that Strategy Nudging diversifies exploration before any policy update is applied, allowing the rollout set to cover a broader range of reasoning modes under the same rollout budget.

3.2 Inter-Intra Group Advantage: Learning to Balance Exploration between Strategies

GRPO estimates advantages by comparing rewards among rollouts conditioned on the same prompt distribution. With Strategy Nudging, however, rollouts are drawn from context-conditioned prompts

$(x_0, z^{(i)})$. A single group baseline therefore entangles reward variation induced by different contexts, distorting the relative advantage assigned to each rollout.

To address this, we propose the *Inter-Intra Group Advantage*, which assigns credit through two complementary signals: an *intra*-context signal, capturing trajectory quality under the same conditioning context, and an *inter*-context signal, capturing the relative reliability of the context itself.

Given sampled rollouts $\{y_i\}_{i=1}^N$ with rewards $r_i = R(x_0, y_i)$, we group them according to their assigned contexts. The set of context groups is defined as

$$\mathcal{G}(x_0) = \text{Unique}(\{z^{(i)}\}) \subseteq \mathcal{C}(x_0) \cup \{\emptyset\}. \quad (8)$$

For each group $g \in \mathcal{G}(x_0)$, we define the index set $I_g = \{i \mid z^{(i)} = g\}$, which partitions all rollouts. We then compute both context-level and global reward baselines:

$$\bar{r}_g = \frac{1}{|I_g|} \sum_{i \in I_g} r_i, \quad \bar{r} = \frac{1}{N} \sum_{i=1}^N r_i. \quad (9)$$

Using these baselines, we define the advantage as:

$$\hat{A}_i = \frac{A_i - \mu_A}{\sigma_A + \delta}, \quad \text{where } A_i = \begin{cases} (r_i - \bar{r}_{z^{(i)}}) + \lambda(\bar{r}_{z^{(i)}} - \bar{r}) & \text{if } z^{(i)} \neq \emptyset, \\ r_i - \bar{r} & \text{if } z^{(i)} = \emptyset. \end{cases} \quad (10)$$

μ_A and σ_A are the mean and standard deviation of $\{A_i\}$, and $\delta > 0$ ensures numerical stability.

Because advantages determine direction of the policy update, they should remain consistent with the underlying rewards while allowing context-level preferences to affect credit assignment.

Proposition 3.1. *Consider two trajectories y and y' sampled from context groups z and z' , with rewards r and r' , respectively. Let $\bar{r}_z$ and $\bar{r}_{z'}$ denote the corresponding context means, and let A and A' denote their advantages. In the binary reward setting, if $\lambda \in [0, 2]$, then:*

$$r > r' \Rightarrow A > A'. \quad (11)$$

Thus, for $\lambda \in [0, 2]$, a higher reward always receives a higher advantage, ensuring consistency with the underlying objective; context only affects the relative ordering among equal-reward trajectories. For equal-reward trajectories, λ controls the context-level preference: $\lambda < 1$ favors successes from lower-reward contexts, encouraging exploration of less typical contexts, whereas $\lambda > 1$ favors successes from higher-reward contexts, emphasizing more reliable contexts. The neutral case $\lambda = 1$ treats equal-reward trajectories identically across contexts; the $\lambda > 1$ case is illustrated in Fig. 2 (a).

3.3 Training objective

Although Strategy Nudging improves exploration by sampling rollouts from context-conditioned prompts $x_1 = (x_0, z)$, the target policy at inference time should operate without external contexts. Therefore, useful trajectories discovered under x_1 must be transferred to the base policy $\pi_\theta(\cdot \mid x_0)$.

To bridge this gap, we introduce an advantage-weighted distillation term following Song et al. [19], which directly updates the policy using trajectories sampled under the context-conditioned input x_1 :

$$\mathcal{L}_{\text{Distill}}(\theta) = -\mathbb{E}_{y \sim \pi_\theta(\cdot \mid x_1)} [\hat{A} \log \pi_\theta(y \mid x_0)]. \quad (12)$$

Unlike standard behavior cloning, this formulation selectively emphasizes trajectories with high normalized advantage, ensuring that only useful behaviors discovered under diverse contexts contribute to the update of $\pi_\theta(\cdot \mid x_0)$.

In parallel, we optimize the reinforcement learning objective on the *context-conditioned policy*:

$$\mathcal{L}_{\text{RL}}(\theta) = -\mathbb{E}_{y \sim \pi_\theta(\cdot \mid x_1)} [\min(r_1 \hat{A}, \text{clip}(r_1, 1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}}) \hat{A})], \quad \text{where } r_1 = \frac{\pi_\theta(y \mid x_1)}{\pi_{\text{old}}(y \mid x_1)}. \quad (13)$$

The final objective combines both terms:

$$\mathcal{L}_{\text{NUGGERL}} = \mathcal{L}_{\text{RL}} + \lambda_{\text{distill}} \mathcal{L}_{\text{Distill}}. \quad (14)$$

Table 1: **Main results** comparing rollout scaling, oracle hinting, and context-based exploration. We report $pass@1$ estimated using 128 rollouts. Best results are represented as **bold** and second best as underline. † indicates additional implementation details; see [Sec. C](#) for details.

Model	Method	Rollouts (N)	AIME24	AIME25	AMC23	MATH500	APEX	Average
Qwen3-4B-Instruct	Base model	–	0.374	0.352	0.653	0.592	0.036	0.402
	GRPO	8	0.444	0.367	0.749	<u>0.668</u>	0.040	0.454
		16	0.454	0.355	0.840	<u>0.655</u>	0.045	0.470
		32	0.451	<u>0.370</u>	0.881	0.674	0.058	0.487
		64	0.415	0.324	0.848	0.641	0.027	0.451
	POPE† [16]	8	<u>0.460</u>	0.337	0.838	0.652	0.048	0.467
	NUDGERL	8	0.482	0.393	<u>0.857</u>	0.660	<u>0.053</u>	0.489
O1mo3-7B-Instruct-SFT	Base model	–	0.134	0.118	0.467	0.384	0.021	0.225
	GRPO	8	0.187	0.159	0.537	0.434	<u>0.025</u>	0.268
		16	0.188	0.176	0.548	0.461	0.023	0.279
		32	0.195	<u>0.176</u>	0.553	<u>0.459</u>	0.024	<u>0.281</u>
		64	0.081	0.053	0.349	0.291	0.027	0.160
	POPE† [16]	8	0.186	0.169	<u>0.558</u>	0.460	0.023	0.279
	NUDGERL	8	<u>0.190</u>	0.179	0.563	0.468	<u>0.025</u>	0.285

This objective induces a complementary learning dynamic. The RL term operates on the context-conditioned policy, improving exploration and reinforcing successful trajectories within each context. In contrast, the distillation term projects these improvements onto the base-prompt policy, enabling cross-context generalization. As a result, the model learns to reproduce effective reasoning strategies without relying on explicit context at inference time. Unlike GRPO in [Eq. 2](#), which samples and optimizes trajectories under the original prompt x_0 , NUDGERL performs RL on context-conditioned rollouts under x_1 while distilling high-advantage trajectories back into the base policy $\pi_\theta(\cdot|x_0)$.

4 Experiments

4.1 Experimental Setup

Baselines. We compare our method against (i) the base model without optimization, which serves as the reference point; (ii) GRPO with increasing rollout budgets, where $N \in \{8, 16, 32, 64\}$, which evaluates naive rollout scaling as a brute-force exploration strategy; and (iii) POPE [16], which augments standard GRPO by appending prefixes of the oracle solution at the end of the base prompt, thereby alleviating the sparse reward signal bottleneck. Further details are provided in [Sec. C](#).

Evaluation Datasets and Metrics. AIME24 and AIME25, 30-problem olympiad-style high-school competitions [13]; AMC23, a 40-problem high-school contest benchmark [12]; the level-5 subset of MATH500, containing 134 difficult MATH problems [4]; and the Apex Shortlist, consisting of 48 advanced competition-style problems [1]. We report $pass@1$, estimated from 128 rollouts using the unbiased estimator of Chen et al. [2]. All solutions are automatically graded using math-verify [6]. Additional details are provided in [Sec. E](#).

Implementation Details. We apply NUDGERL to Qwen3-4B-Instruct-2507 [21] and O1mo3-7B-Instruct-SFT [14] using DAP0-17k-Processed as a training set [24]. To construct the pool of contexts, we used gpt-4o-mini [15] to generate two strategy-level contexts per problem (e.g., Pythagorean theorem), and used them without additional verification (i.e., $\forall x \in \mathcal{D}, |\mathcal{C}(x)| = 2$). For the POPE baseline, oracle solutions were generated using DeepSeek Reasoner v3.2 [9]. We provide additional optimization details in [Sec. D](#).

4.2 Main Results

NudgeRL matches larger-budget GRPO with fewer rollouts. As shown in [Tab. 1](#), NUDGERL achieves the best average performance on both models while using only 8 rollouts per prompt. On Qwen3-4B-Instruct-2507, NUDGERL reaches 0.489 average $pass@1$, slightly outperforming the best GRPO result at 32 rollouts (0.487) and surpassing GRPO at 64 rollouts (0.451) with an $8\times$

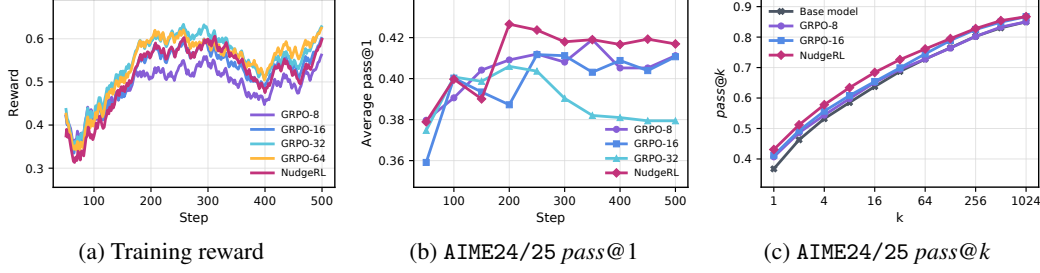


Figure 3: **Training dynamics and evaluation performance** on Qwen3-4B-Instruct. (a) EMA-smoothed training reward with decay factor 0.99. (b, c) Average $pass@1$ and $pass@k$ on AIME24/25, estimated from 64 sampled rollouts using the unbiased estimator.

smaller rollout budget. On `01mo3-7B-Instruct-SFT`, NUDGERL likewise improves over the best GRPO result, achieving 0.285 compared to 0.281 at 32 rollouts. These results indicate that larger rollout budgets alone are not sufficient: GRPO improves up to $N = 32$ but degrades at $N = 64$ on both models, suggesting instability under brute-force rollout scaling. In contrast, NUDGERL achieves stronger performance by improving the quality of exploration through Strategy Nudging, rather than relying on more sampled rollouts.

Comparison with oracle-prefix method. We also compare with POPE [16], which augments GRPO by generating rollouts conditioned on the oracle solution prefixes. Unlike baselines relying on expensive, unscalable oracle hints [16] or text feedback [19], our approach ensures scalable diversity. We use a lightweight LLM (e.g., `gpt-4o-mini`) to cheaply generate unverified strategy-level contexts that induce multiple reasoning directions. Despite this weaker supervision, our method consistently outperforms oracle-guided baselines, demonstrating that structured exploration over diverse strategies is more effective than injecting narrow, privileged solution signals.

4.3 Efficient Coverage of Diverse Reasoning Modes

As discussed in Sec. 3.1, relying solely on scaling the rollout budget suffers from severe sample inefficiency when discovering long-tail, low-probability reasoning modes. This is because naive rollout scaling repeatedly allocates computation to dominant trajectories. To empirically investigate how Strategy Nudging overcomes this exploration bottleneck and improves sample efficiency, we compare the training dynamics of NUDGERL against GRPO under progressively larger rollout budgets. We evaluate the model for every 50 training steps on the combined AIME24 and AIME25 benchmark by sampling 64 rollouts per problem and estimating $pass@1$ and $pass@8$.

As shown in Fig. 3b, NUDGERL improves $pass@1$ faster than GRPO variants and remains the strongest method throughout most of training. By 200 steps, NUDGERL exceeds 0.42 $pass@1$ on AIME24/25, while GRPO variants remain around or below 0.41 and show slower or less stable gains as the rollout budget increases. This suggests that Strategy Nudging improves sample efficiency by exposing useful reasoning trajectories earlier, rather than merely increasing sampled rollouts. Enlarging the number of samples (k) further validates this trend under the same training rollout budget. As shown in Fig. 3c, NUDGERL consistently outperforms GRPO-8 across the full k range, which indicates that Strategy Nudging improves inference-time sample efficiency, requiring fewer generated solutions to reach the same level of $pass@k$.

4.4 Case Study

To examine the source of performance gains in NUDGERL, we analyze one AIME25 problem where the NUDGERL-trained model successfully sampled correct trajectories, while the GRPO-trained model entirely failed. We sampled 32 rollouts and categorized their dominant reasoning strategies.

As shown in Fig. 4, both models predominantly relied on *coordinate geometry*. However, the GRPO-trained model additionally explored ineffective strategies such as *symmetry assumptions* and *area decomposition*, which consistently resulted in truncated solutions, causing all 32 trajectories to fail. While GRPO sampled the *shoelace formula* strategy only once, NUDGERL substantially increased its frequency and successfully exploited it to generate correct trajectories.

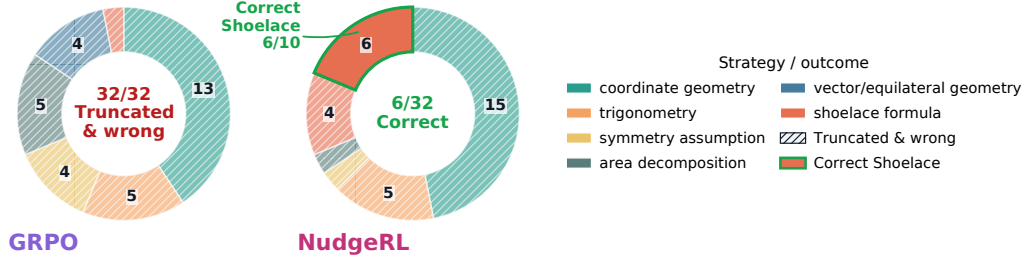


Figure 4: **NUDGERL internalizes effective test-time strategies.** Across 32 rollouts on a AIME25 problem, GRPO yields only incorrect and truncated trajectories. Conversely, NUDGERL produces 6 correct solutions using the shoelace formula.

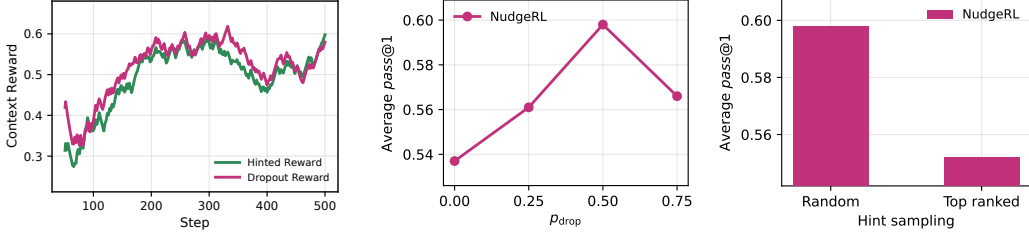


Figure 5: Training dynamics. We report time-weighted EMA reward mean (0.99) with and without context.

(a) Ablation on p_{drop} (b) Ablation on sampling
Figure 6: Ablation results on sampling. We report Average pass@1 estimated using 128 rollouts on AIME24/25, AMC23, MATH500.

This behavior highlights the complementary roles of our framework: *Strategy Nudging* exposes rare but effective reasoning modes such as the shoelace-formula strategy, while the *Inter-Intra Group Advantage* reinforces and exploits such reliable strategies once discovered. Details are in [Sec. F](#).

4.5 Effect of Contexts during training

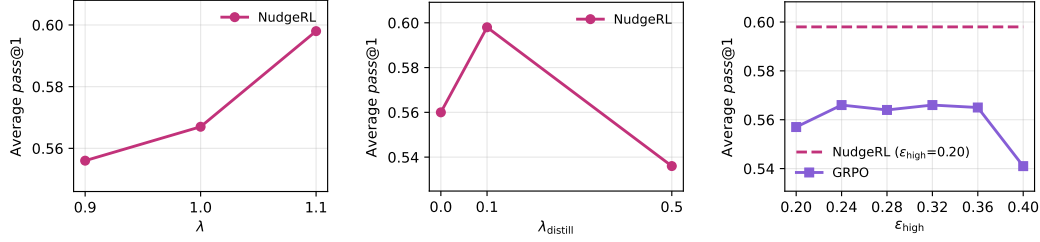
We also report the dropout reward mean ($\mathbb{E}_i[r(x_0, y^{(i)}) \mid z^{(i)} = \emptyset]$) and the hinted reward mean ($\mathbb{E}_i[r(x_0, y^{(i)}) \mid z^{(i)} \neq \emptyset]$) during training of Qwen3-4B-Instruct-2507 with NUDGERL. As shown in [Fig. 5](#), both rewards improve together throughout training, suggesting that trajectories discovered under context-conditioned exploration are successfully transferred to the base policy through the distillation objective. Interestingly, the dropout reward occasionally exceeds the hinted reward during training. This contrasts with prior feasibility-oriented methods based on privileged information [8, 16, 19, 27]. In ours, primary role of context is not to directly simplify the problem, but to induce diverse reasoning trajectories that can later be internalized by the context-free policy.

4.6 Underlying Mechanism of NUDGERL

To further understand the source of performance gains in NUDGERL, we conduct a series of controlled experiments using Qwen3-4B-Instruct-2507 [21] on a subset of benchmarks.

p_{drop} Ablation. As shown in [Fig. 6a](#), a moderate dropout rate ($p_{\text{drop}} = 0.5$) consistently yields the best performance across benchmarks. Context dropout plays a dual role: it enables exploration beyond fixed contexts by occasionally reverting to the base prompt, while also stabilizing group-wise statistics through a more balanced sample distribution. When $p_{\text{drop}} = 0$, exploration is restricted to predefined contexts, whereas large values diminish the influence of context forcing. These results suggest that maintaining a balanced mixture of context-conditioned and context-free samples is important for achieving both diverse exploration and stable optimization.

Hint Sampling. We study how the quality of sampled contexts affects performance by comparing two strategies: *random* sampling and *top-ranked* selection. In the top-ranked setting, we first generate a pool of five candidate contexts, and then select the two that yield the largest improvement in pass@16 for each problem, as measured by oracle evaluation.



(a) Ablation results on λ (b) Ablation results on λ_{distill} (c) ϵ_{high} scaling results
Figure 7: Ablation on learning and ϵ_{high} scaling results. We report Average *pass@1* estimated using 128 rollouts on AIME24/25, AMC23, MATH500 dataset.

As shown in Fig. 6b, random sampling consistently outperforms top-ranked selection in terms of *pass@1*. While top-ranked contexts ensure more correctness, they tend to concentrate on a narrow set of reasoning strategies. In contrast, random sampling induces a broader distribution over plausible trajectories, resulting in more effective exploration under limited rollout budgets.

These results suggest that, within our framework, the primary role of context is not to provide the single best hint, but to promote diversity in reasoning. Consequently, simple random sampling is not only sufficient, but also preferable for scalable and effective context-based exploration.

Exploration-Exploitation trade-off via λ . Fig. 7a presents the effect of varying λ , where $\lambda = 1.1$ achieves the best performance. This trend aligns with our Proposition 3.1 in Sec. 3.2. Since strategy nudging already ensures sufficient diversity at the sampling stage, increasing λ does not hinder exploration across contexts. Instead, it strengthens exploitation within each problem by prioritizing trajectories from more reliable contexts. This leads to more consistent learning of high-quality solutions per instance, explaining the observed performance gains at $\lambda = 1.1$.

Distillation Coefficient. As shown in Fig. 7b, removing the distillation term ($\lambda_{\text{distill}} = 0$) results in a clear performance drop, indicating that explicitly transferring context-discovered trajectories to the base policy is essential. However, overly large values also degrade performance, likely due to over-constraining the policy toward sampled trajectories. A moderate coefficient ($\lambda_{\text{distill}} = 0.1$) achieves the best results, suggesting that distillation should complement the underlying RL objective.

4.7 Comparison with ϵ_{high} scaling.

We further compare our algorithm with decoupled clipping [24]: $\text{clip}(r, 1 - \epsilon_{\text{low}}, 1 + \epsilon_{\text{high}})$, where ϵ_{high} controls the strength of policy updates by amplifying the contribution of successful trajectories. Increasing ϵ_{high} therefore allows more aggressive policy updates toward positive-advantage trajectories. As shown in Fig. 7c, increasing ϵ_{high} generally improves GRPO performance in the moderate regime used in prior works [18, 24]. However, our method with $\epsilon_{\text{high}} = 0.2$ consistently outperforms GRPO across the entire scaling range from moderate to extreme values. This suggests that improving exploration quality is more effective than simply increasing the magnitude of stochastic policy updates. Additionally, under the more extreme scaling adopted in recent RLVR settings [10], GRPO sharply deteriorates at $\epsilon_{\text{high}} = 0.4$. We argue that this degradation highlights a limitation of purely stochastic distribution-level exploration: increasing update magnitude alone provides little control over *what* is explored.

The complete results of the evaluation are given in the Sec. G.

5 Conclusion

In this work, we introduced NUDGERL, a framework for structured exploration in RLVR. Our approach leverages *Strategy Nudging* to induce diverse reasoning trajectories by sampling from lightweight, strategy-level context-conditioned distributions, and learns from them via distillation augmented RL objective. Empirically, NUDGERL achieves superior performance compared to GRPO using up to $8\times$ larger rollout budgets, and further outperforms oracle prefix-based baselines across models.

Limitations & Future Work A practical consideration of NUDGERL is the cost of generating strategy-level contexts. However, this is an *offline* process performed once prior to training, using a lightweight LLM (e.g., gpt-4o-mini), and the resulting contexts can be reused across training runs without additional overhead. A more fundamental limitation lies in how contexts are generated independently of the model being trained. The benefit of Context Forcing stems from inducing trajectories that are unlikely under the current policy. As training progresses, however, a fixed context pool may become less informative as the policy adapts. A promising direction for future work is *model-adaptive context generation*, which dynamically constructs contexts tailored to the current policy’s blind spots, potentially yielding more consistent exploration gains throughout training.

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A Related Work

A.1 Reinforcement Learning with Verifiable Rewards

Reinforcement Learning with Verifiable Rewards (RLVR) has emerged as a scalable paradigm for improving the reasoning abilities of large language models[9, 18, 20, 24]. By leveraging automatically verifiable signals, such as exact answers in mathematics or test-case correctness in code generation, RLVR enables effective policy optimization without dense human supervision.

A representative approach is *Group-Relative Policy Optimization* (GRPO) [18], which replaces value function estimation with group-wise comparisons among sampled rollouts, deriving advantages from relative reward differences within each group. Building on this formulation, subsequent work has introduced improvements such as decoupled clipping [24] and alternative normalization strategies [11] to enhance training stability.

These methods have been successfully applied across a range of reasoning tasks and model scales [9, 10], establishing RLVR as a standard post-training approach for LLMs. However, their effectiveness fundamentally depends on exploration: the policy can only improve on trajectories it has already sampled. As a result, insufficient exploration directly limits learning, making it a key bottleneck in RLVR. We next examine how prior work addresses this challenge.

A.2 Exploration in RLVR

A straightforward approach to improving exploration is to scale the number of sampled rollouts. Prior work has shown that such rollout scaling can significantly improve performance by reducing the probability mass of un-sampled region[5]. However, this approach is computationally expensive and often impractical at scale.

More commonly, recent methods attempt to encourage exploration through objective design, such as entropy regularization [3, 26] or decoupled clipping [24]. While these approaches can steer the update toward exploration, they do not guarantee that useful or rare modes are actually sampled during training. In other words, shaping the distribution does not necessarily ensure coverage of meaningful trajectories, leaving exploration fundamentally limited.

Moreover, such distribution-level exploration is inherently stochastic and unconstrained, which can perturb the policy in semantically undesirable directions. Increasing entropy or aggressively reweighting probabilities may encourage the model to explore low-probability regions, but without any structural guidance, this often leads to incoherent or unproductive trajectories rather than meaningful reasoning strategies. As a result, these approaches lack control over *how* the policy explores, and fail to provide structured, strategy-level exploration that targets diverse and semantically valid modes of reasoning.

A.3 Usage of Privileged Information

Another key limitation of widely used group-based advantage methods, such as GRPO [18], is that they rely on relative comparisons within a group of rollouts. When all samples in a group are either correct or incorrect, these methods fail to provide informative learning signals.

To address this issue, recent works have introduced privileged information to assist the policy [8, 16, 19, 27], often in the form of oracle prefixes or intermediate solutions. These approaches improve the feasibility of solving hard problems by enabling the model to generate successful trajectories that would otherwise be unreachable.

However, such methods come with several limitations. First, privileged information is often difficult to scale, especially when it relies on oracle solutions or expensive annotations [16, 27]. Second, the mechanism by which the model internalizes this information and performs well without it at test time remains unclear [8]. Third, many approaches assume multi-turn or interactive settings [19], which may not align with standard single-turn RLVR setups.

More importantly, existing work primarily focuses on improving the feasibility of generating correct trajectories on difficult problems. In contrast, our work targets a complementary challenge: improving the diversity of exploration, even when successful trajectories are already attainable.

B Details on Strategy Nudging

Strategy Generating Prompt. We use gpt-4o-mini to generate keyword-level hints for each problem. For the main experiments, we generate two hints per problem, while in the top-ranked setting, we first generate five candidate hints and select a subset based on oracle evaluation.

The exact prompt used for hint generation is as follows:

```
f"""Given the following math problem, generate {num_hints} different keyword hints
that would help solve it.
Each hint should be a specific mathematical concept, theorem, or technique (e.g.,
"Ceva's theorem", "Lifting the exponents", "Triangle inequality").

Problem:
{problem}

Please provide exactly {num_hints} hints in the following format (one hint per
line, numbered):
1. [Hint 1]
2. [Hint 2]
...
{num_hints}. [Hint {num_hints}]

Make sure each hint is a distinct mathematical concept or theorem."""
```

Strategy Nudging prompt. Given a problem and an optional hint, we construct prompts that encourage the model to follow a specific reasoning strategy. The model is instructed to explicitly separate its reasoning process and final answer using predefined delimiters.

```
reasoning_start = "<start_working_out>"
reasoning_end = "<end_working_out>"
solution_start = "<SOLUTION>"
solution_end = "</SOLUTION>"

system_prompt = f"""You are given a problem.
Think about the problem and provide your working out.
Place it between {reasoning_start} and {reasoning_end}.
Then, provide your solution between {solution_start}{solution_end}"""

def build_messages(problem: str, system_prompt: str, hint: str | None = None)
-> list[dict[str, str]]:
    context_block = ""
    if hint:
        context_block = (
            "Context (exploration condition):\n"
            f"- Use this hint/approach: {hint}\n\n"
            "Important:\n"
            "- Follow this approach as your primary strategy.\n\n"
        )
    user_content = (
        "Problem:\n"
        f"{problem}\n\n"
        f"{context_block}"
        "Solve this step by step and provide your final numerical answer at the"
        "end."
    )
    return [
        {"role": "system", "content": system_prompt},
        {"role": "user", "content": user_content},
    ]
```

Effect of Strategy Nudging. To evaluate the effect of Strategy Nudging, we sample 8 rollouts from Qwen3-4B-Instruct-2507 [21] on 200 problems from DAP0-17k-Processed [24], both with and without Strategy Nudging, and analyze the resulting rollout diversity via LLM-as-a-judge.

LLM-as-a-judge prompt. To analyze the diversity of generated rollouts, we employ an LLM-as-a-judge using gpt-4o-mini to cluster solutions based on their underlying reasoning strategies and count the number of distinct solution modes. Given a problem and a set of rollouts, the model is instructed to identify the number of *conceptually distinct* solution approaches, while ignoring superficial differences such as phrasing or minor computational variations.

```
prompt = (f"Problem:\n{problem_text}\n\n"
          f"Here are {len(rollouts)} student solutions to this problem:\n"
          f"{formatted_rollouts}\n"
          f"Analyze these solutions and determine how many *conceptually"
          f"distinct* solution strategies are used across them.\n"
          f"Ignore minor calculation differences or phrasing variations. Focus"
          f"on the core mathematical approach.\n"
          f"Provide the answer in the following format: 'Distinct Strategies: X'"
          f"where X is the integer count.\n"
          f"Then briefly list the distinct strategies identified.")
messages = [
    {"role": "system", "content": "You are an expert math teacher evaluating"
    the diversity of student solution methods."},
    {"role": "user", "content": prompt}
]
```

C Details on Baselines

Rollout Scaling in GRPO. For controlled experiments, we scale the number of rollouts per prompt while adjusting the gradient accumulation steps and generation batch size accordingly, as summarized in Tab. 2. This ensures that the total optimization dynamics remain comparable across different rollout settings.

Implementing POPE [16]. To compare our method with oracle prefix-based approaches, we implement our own version of POPE [16]. We follow the original paper in using the same prompt format and dataset mixture (i.e., with and without privileged information). Since the length of oracle solutions varies across prior works [16, 27], we standardize this by truncating the oracle solution to 15% of its full length when used as a prefix.

Example of Generated Contexts. We provide an illustrative example of the strategy-level contexts used in our method. These contexts are lightweight, keyword-level hints that do not directly solve the problem, but instead steer the model toward distinct reasoning modes. Importantly, they are not intended to provide intermediate steps or solutions, but rather to act as high-level inductive biases that diversify exploration.

Problem

Let S be the set of triples (a, b, c) of non-negative integers such that $a + b + c$ is even. The value of the sum

$$\sum_{(a,b,c) \in S} \frac{1}{2^a 3^b 5^c}$$

can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Compute $m + n$.

Oracle solution:

We compute the sum

$$\sum_{(a,b,c) \in S} \frac{1}{2^a 3^b 5^c},$$

where S consists of non-negative integer triples (a, b, c) with $a + b + c$ even. Using the indicator

$$\mathbf{1}_{a+b+c \text{ even}} = \frac{1}{2} \left(1 + (-1)^{a+b+c} \right),$$

we write

$$\sum_{(a,b,c) \in S} \frac{1}{2^a 3^b 5^c} = \frac{1}{2} \left(\sum_{a,b,c \geq 0} \frac{1}{2^a 3^b 5^c} + \sum_{a,b,c \geq 0} \frac{(-1)^{a+b+c}}{2^a 3^b 5^c} \right).$$

The first sum factors as

$$\sum_{a \geq 0} \frac{1}{2^a} \cdot \sum_{b \geq 0} \frac{1}{3^b} \cdot \sum_{c \geq 0} \frac{1}{5^c} = \frac{1}{1 - \frac{1}{2}} \cdot \frac{1}{1 - \frac{1}{3}} \cdot \frac{1}{1 - \frac{1}{5}} = 2 \cdot \frac{3}{2} \cdot \frac{5}{4} = \frac{15}{4}.$$

The second sum factors as

$$\sum_{a \geq 0} \frac{(-1)^a}{2^a} \cdot \sum_{b \geq 0} \frac{(-1)^b}{3^b} \cdot \sum_{c \geq 0} \frac{(-1)^c}{5^c} = \frac{1}{1 + \frac{1}{2}} \cdot \frac{1}{1 + \frac{1}{3}} \cdot \frac{1}{1 + \frac{1}{5}} = \frac{2}{3} \cdot \frac{3}{4} \cdot \frac{5}{6} = \frac{5}{12}.$$

Thus, the total sum is

$$\frac{1}{2} \left(\frac{15}{4} + \frac{5}{12} \right) = \frac{1}{2} \cdot \frac{50}{12} = \frac{25}{12}.$$

Since $\frac{25}{12}$ is in lowest terms, we have $m = 25$, $n = 12$, and $m + n = 37$.

Strategy-level contexts(ours):

"Generating functions", "Parity of sums"

D Training Detail

Framework. We used TRL [22] for implementing baselines and our algorithm.

Hyperparameters. The hyperparameters we used in training are given in Tab. 2.

Compute resources. For all experiments, we used NVIDIA H200 140GB GPUs.

E Details on Evaluation

During evaluation, all hyperparameters are kept identical to Tab. 2, except for the temperature, which is set to 0.7.

F Details on Case study

In this section, we provide qualitative examples from the case study presented in Fig. 4.

The GRPO-trained model predominantly relied on coordinate geometry combined with heuristic symmetry assumptions and case-by-case area decomposition. Although these approaches occasionally progressed toward partial solutions, they frequently resulted in excessively long derivations and truncated outputs before reaching the final answer.

In contrast, NUDGERL exploited the *shoelace-formula* strategy, which directly computes polygon areas from vertex coordinates. This strategy produced substantially shorter and more reliable reasoning trajectories, enabling successful completion within the generation budget.

Table 2: Hyperparameters for training.

Parameter	Value
LoRA rank	32
Max prompt length	2,048
Max completion length	6,144
RL steps	500
Batch size	4
Rollouts per prompt	N
Gradient accumulation steps	$\frac{N}{8}$
Generation batch size	N
Temperature	1
Min- p	0.0
Top- p	0.95
Top- k	-1
Learning rate	2×10^{-5}
LR scheduler	cosine
Weight decay	0.001
Warmup ratio	0.05
Optimizer	AdamW (8-bit)
KL coefficient β	0
Epsilon low	0.2
Epsilon high	0.2 (unless specified)
λ	1.1
λ_{distill}	0.1
p_{drop}	0.5
Random seed	42

Problem

Let $\triangle ABC$ be a right triangle with $\angle A = 90^\circ$ and $BC = 38$. There exist points K and L inside the triangle such that

$$AK = AL = BK = CL = KL = 14.$$

The area of the quadrilateral $BKLC$ can be expressed as $n\sqrt{3}$ for some positive integer n . Find n .

GRPO solution

<start_working_out>

We are given a right triangle $\triangle ABC$ with $\angle A = 90^\circ$, and hypotenuse $BC = 38$. There are two points K and L inside the triangle such that:

$$AK = AL = BK = CL = KL = 14$$

We are to find the area of quadrilateral $BKLC$, which is expressed as $n\sqrt{3}$, and determine n .

Step 1: Understand the configuration.

Given that $AK = AL = 14$, and $BK = CL = 14$, and $KL = 14$, all these segments are equal to 14. So, we have:

- $AK = AL = 14$: So point K and L are both at distance 14 from A - $BK = 14$, $CL = 14$: So both K and L are at distance 14 from B and C , respectively - $KL = 14$

So, K lies at intersection of: - Circle centered at A with radius 14 - Circle centered at B with radius 14

Similarly, L lies at intersection of: - Circle centered at A with radius 14 - Circle centered at C with radius 14

And additionally, $KL = 14$

So both K and L lie on the intersection of circle centered at A and circles centered at B and C , respectively.

Moreover, $KL = 14$, so the distance between K and L is 14.

Step 2: Use symmetry.

Since the triangle is right-angled at A , and the construction is symmetric (both K and L are symmetric with respect to A , and distances to B and C are equal), it is likely that the triangle is isosceles right triangle.

Suppose $AB = AC$. Then $\triangle ABC$ is an isosceles right triangle with legs $AB = AC = x$, and hypotenuse $BC = x\sqrt{2} = 38$

Then:

$$x\sqrt{2} = 38 \Rightarrow x = \frac{38}{\sqrt{2}} = 19\sqrt{2}$$

So $AB = AC = 19\sqrt{2}$

Now, consider the point K such that $AK = BK = 14$

Similarly, L such that $AL = CL = 14$

Because of symmetry, if the triangle is symmetric about the angle bisector (the line $y = x$ if we place A at origin), then K and L are symmetric points with respect to this line.

So K lies in one region, L in the mirror region.

Then $KL = 14$, and we can compute the distance between such symmetric points.

Let's try to place coordinates.

Step 3: Coordinate setup.

Place point A at origin $(0, 0)$

Let $B = (b, 0)$, $C = (0, c)$

Since $\angle A = 90^\circ$, this is valid.

We are told $BC = 38$

By distance formula:

$$BC^2 = (b - 0)^2 + (0 - c)^2 = b^2 + c^2 = 38^2 = 1444$$

We also have points K and L such that:

- $AK = 14 \rightarrow K$ lies on circle $x^2 + y^2 = 196$ - $BK = 14 \rightarrow (x - b)^2 + y^2 = 196$ - $CL = 14 \rightarrow x^2 + (y - c)^2 = 196$ - $AL = 14 \rightarrow x^2 + y^2 = 196$ - $KL = 14$

So both K and L lie on the circle centered at A with radius 14.

Let's find coordinates of K : intersection of circles: - $x^2 + y^2 = 196$ (1) - $(x - b)^2 + y^2 = 196$ (2)

Subtract (1) from (2):

$$(x - b)^2 - x^2 = 0 \Rightarrow x^2 - 2bx + b^2 - x^2 = -2bx + b^2 = 0 \Rightarrow -2bx + b^2 = 0 \Rightarrow x = \frac{b}{2}$$

Plug into (1):

$$\left(\frac{b}{2}\right)^2 + y^2 = 196 \Rightarrow \frac{b^2}{4} + y^2 = 196 \Rightarrow y^2 = 196 - \frac{b^2}{4}$$

Similarly, for point L , since $CL = 14$, and $AL = 14$, by symmetry, we get:

$x^2 + (y - c)^2 = 196$, and $x^2 + y^2 = 196$

Subtract: $(y - c)^2 - y^2 = 0 \Rightarrow -2cy + c^2 = 0 \Rightarrow y = \frac{c}{2}$

Then from $x^2 + y^2 = 196$: $x^2 + \left(\frac{c}{2}\right)^2 = 196 \Rightarrow x^2 = 196 - \frac{c^2}{4}$

So coordinates:

- $K = \left(\frac{b}{2}, y_K\right)$, where $y_K^2 = 196 - \frac{b^2}{4}$ - $L = \left(x_L, \frac{c}{2}\right)$, where $x_L^2 = 196 - \frac{c^2}{4}$

Now, $KL = 14$

So compute distance between K and L :

$$KL^2 = \left(\frac{b}{2} - x_L\right)^2 + \left(y_K - \frac{c}{2}\right)^2 = 196$$

GRPO Solution (Cont.)

This seems messy unless $b = c$, i.e., isosceles right triangle.

Try assuming $b = c$

Then $b^2 + c^2 = 2b^2 = 1444 \Rightarrow b^2 = 722 \Rightarrow b = \sqrt{722} = \sqrt{7 \cdot 22}$

Now, for K : $x_K = \frac{b}{2}$, $y_K^2 = 196 - \frac{b^2}{4} = 196 - \frac{722}{4} = 196 - 180.5 = 15.5$

Similarly, $x_L^2 = 196 - \frac{722}{4} = 15.5$, so $x_L = \sqrt{15.5}$

So $K = \left(\frac{b}{2}, \sqrt{15.5}\right)$, $L = \left(\sqrt{15.5}, \frac{b}{2}\right)$

Now compute KL^2 :

$$KL^2 = \left(\frac{b}{2} - \sqrt{15.5}\right)^2 + \left(\sqrt{15.5} - \frac{b}{2}\right)^2 = 2\left(\frac{b}{2} - \sqrt{15.5}\right)^2$$

Set $d = \frac{b}{2}$, then $b^2 = 722 \Rightarrow d^2 = \frac{722}{4} = 180.5$

Then $KL^2 = 2(d - \sqrt{15.5})^2$

Compute $d = \sqrt{180.5} \approx 13.43$

$\sqrt{15.5} \approx 3.937$

Then $d - \sqrt{15.5} \approx 13.43 - 3.937 = 9.493$

Then $KL^2 \approx 2 \times (9.493)^2 \approx 2 \times 90.1 \approx 180.2$, which is not 196.

But we need $KL = 14$, so $KL^2 = 196$

So this value is too small — contradiction.

Therefore, in the isosceles case, $KL^2 \approx 180 < 196$, so not satisfying the condition.

So the triangle is not isosceles?

But the symmetry suggests it might be.

Alternatively, maybe the configuration implies symmetry, and the triangle is isosceles, but our calculation is wrong.

Wait — perhaps we assumed the wrong point.

Wait: we assumed that K is equidistant from A and B , so lies at $x = b/2$

But is that correct?

Yes — from the two equations:

$$x^2 + y^2 = 196$$

$$(x - b)^2 + y^2 = 196$$

$$\text{Subtract: } -2bx + b^2 = 0 \Rightarrow x = b/2$$

Yes.

Similarly, for L , $y = c/2$

So coordinates are correct.

$$\text{Then } KL^2 = (b/2 - x_L)^2 + (y_K - c/2)^2$$

$$\text{With } x_L^2 = 196 - c^2/4, y_K^2 = 196 - b^2/4$$

So define:

$$\text{Let } u = b/2, v = c/2$$

$$\text{Then } b = 2u, c = 2v$$

$$\text{Then } b^2 + c^2 = 4u^2 + 4v^2 = 1444 \Rightarrow u^2 + v^2 = 361$$

$$\text{Now, } y_K^2 = 196 - u^2, \text{ so } y_K = \sqrt{196 - u^2}$$

$$\text{Similarly, } x_L^2 = 196 - v^2, \text{ so } x_L = \sqrt{196 - v^2}$$

Then:

$$KL^2 = (u - \sqrt{196 - v^2})^2 + (\sqrt{196 - u^2} - v)^2 = 196$$

This is symmetric in u and v , so likely $u = v$

Try $u = v$

$$\text{Then } u^2 + v^2 = 2u^2 = 361 \Rightarrow u^2 = 180.5$$

$$\text{Then } u = \sqrt{180.5} \approx 13.43$$

$$\text{Then } \sqrt{196 - u^2} = \sqrt{196 - 180.5} = \sqrt{15.5} \approx 3.937$$

$$\text{Then } KL^2 = 2(u - \sqrt{15.5})^2 \approx 2(13.43 - 3.937)^2 = 2(9.493)^2 \approx 2 \times 90.1 = 180.2 \neq 196$$

Still too small.

So even with symmetry, $KL^2 < 196$

But we need $KL^2 = 196$

So contradiction.

Therefore, the triangle is not symmetric?

But then why are $AK = AL = BK = CL = 14$?

There is symmetry: K is symmetric to L in some way?

Wait — maybe K and L are symmetric with respect to the angle bisector.

But unless the triangle is isosceles, such symmetry is not possible.

Alternatively, perhaps the points K and L are such that triangle AKL is equilateral?

Given $AK = AL = KL = 14$, so triangle AKL is equilateral!

Yes! This is key.

We are told $AK = AL = KL = 14$, so $\triangle AKL$ is equilateral.

So $\angle KAL = 60^\circ$

But $\angle A = 90^\circ$, so that's possible.

So point K and L are at distance 14 from A , and from each other, forming an equilateral triangle.

GRPO Solution (Cont.)

Also, $BK = 14$, $CL = 14$

So K lies on intersection of circles: center A , radius 14 and center B , radius 14

Similarly, L lies on intersection of circles: center A , radius 14 and center C , radius 14

So we can find coordinates.

Let's proceed with coordinate geometry.

Let $A = (0, 0)$, $B = (c, 0)$, $C = (0, b)$, so $AB = c$, $AC = b$, $BC = \sqrt{b^2 + c^2} = 38$

So $b^2 + c^2 = 1444$

Now, point K lies at intersection of: $x^2 + y^2 = 196$ (distance from A) - $(x - c)^2 + y^2 = 196$ (distance from B)

Subtract: $(x - c)^2 - x^2 = 0 \Rightarrow -2cx + c^2 = 0 \Rightarrow x = \frac{c}{2}$

Then $(\frac{c}{2})^2 + y^2 = 196 \Rightarrow y^2 = 196 - \frac{c^2}{4}$

Similarly, point L lies on: $x^2 + y^2 = 196$ - $x^2 + (y - b)^2 = 196$

Subtract: $(y - b)^2 - y^2 = -2by + b^2 = 0 \Rightarrow y = \frac{b}{2}$

Then $x^2 + (\frac{b}{2})^2 = 196 \Rightarrow x^2 = 196 - \frac{b^2}{4}$

So: $K = (\frac{c}{2}, y_K)$, $y_K = \sqrt{196 - \frac{c^2}{4}}$ - $L = (x_L, \frac{b}{2})$, $x_L = \sqrt{196 - \frac{b^2}{4}}$

Now, $KL = 14$, so:

$$KL^2 = \left(\frac{c}{2} - x_L\right)^2 + \left(y_K - \frac{b}{2}\right)^2 = 196$$

Let's define: $u = \frac{c}{2}$, $v = \frac{b}{2}$

Then $c = 2u$, $b = 2v$, and $b^2 + c^2 = 4u^2 + 4v^2 = 1444 \Rightarrow u^2 + v^2 = 361$

Now: $y_K = \sqrt{196 - u^2}$ - $x_L = \sqrt{196 - v^2}$

Then:

$$KL^2 = (u - \sqrt{196 - v^2})^2 + (\sqrt{196 - u^2} - v)^2 = 196$$

Let's expand this expression:

First term: $(u - \sqrt{196 - v^2})^2 = u^2 - 2u\sqrt{196 - v^2} + (196 - v^2)$

Second term: $(\sqrt{196 - u^2} - v)^2 = (196 - u^2) - 2v\sqrt{196 - u^2} + v^2$

Add both:

Total:

$$\begin{aligned} &u^2 + 196 - v^2 + 196 - u^2 + v^2 - 2u\sqrt{196 - v^2} - 2v\sqrt{196 - u^2} \\ &= 392 - 2u\sqrt{196 - v^2} - 2v\sqrt{196 - u^2}. \end{aligned}$$

Set equal to 196:

$$392 - 2u\sqrt{196 - v^2} - 2v\sqrt{196 - u^2} = 196$$

$$2u\sqrt{196 - v^2} + 2v\sqrt{196 - u^2} = 296$$

$$u\sqrt{196 - v^2} + v\sqrt{196 - u^2} = 148.$$

Now, recall that $u^2 + v^2 = 361$

Let $S = u\sqrt{196 - v^2} + v\sqrt{196 - u^2} = 148$

We can try to assume symmetry: $u = v$

Then $2u^2 = 361 \Rightarrow u^2 = 180.5$

Then $u = \sqrt{180.5} \approx 13.43$

Then $\sqrt{196 - u^2} = \sqrt{196 - 180.5} = \sqrt{15.5} \approx 3.937$

Then $S = u \cdot \sqrt{15.5} + v \cdot \sqrt{15.5} = 2u \cdot \sqrt{15.5} \approx 2 \times 13.43 \times 3.937 \approx 2 \times 52.9 \approx 105.8 \neq 148$

Too small.

So not symmetric.

We need to solve:

$$u^2 + v^2 = 361$$

$$u\sqrt{196 - v^2} + v\sqrt{196 - u^2} = 148$$

Try to make a substitution.

Let $a = u^2$, $b = v^2$, so $a + b = 361$

Then $u = \sqrt{a}$, $v = \sqrt{b}$

So the equation becomes:

$$\sqrt{a}\sqrt{196 - b} + \sqrt{b}\sqrt{196 - a} = 148$$

Note: $196 - b = 196 - (361 - a) = 196 - 361 + a = a - 165$

Similarly, $196 - a = b - 165$

So:

$$\sqrt{a(a - 165)} + \sqrt{b(b - 165)} = 148$$

But $b = 361 - a$, so:

$$\sqrt{a(a - 165)} + \sqrt{(361 - a)(361 - a - 165)} = 148$$

Simplify:

$$\sqrt{a(a - 165)} + \sqrt{(361 - a)(196 - a)} = 148$$

GRPO Solution (Cont.)

Now, define function:

$$f(a) = \sqrt{a(a-165)} + \sqrt{(361-a)(196-a)}$$

We want $f(a) = 148$

Domain: $a > 165$, and $361 - a > 0 \Rightarrow a < 361$, and $196 - a > 0 \Rightarrow a < 196$

So $165 < a < 196$

Try $a = 180$

Then:

$$- a(a-165) = 180 \times 15 = 2700, \sqrt{2700} \approx 51.96 - (361-180)(196-180) = (181)(16) = 2896, \sqrt{2896} \approx 53.82 - \text{Sum} \approx 51.96 + 53.82 = 105.78 \rightarrow \text{too small}$$

Try $a = 190$

$$- a(a-165) = 190 \times 25 = 4750, \sqrt{4750} \approx 68.92 - (361-190)(196-190) = (171)(6) = 1026, \sqrt{1026} \approx 32.03 - \text{Sum} \approx 68.92 + 32.03 = 100.95 \rightarrow \text{even smaller}$$

Wait — it's decreasing?

Wait, when a increases, first term increases, second decreases.

But at $a = 165$, first term is 0, second term: $(361-165)(196-165) = (196)(31) = 6076, \sqrt{6076} \approx 77.95$

At $a = 196$, first term: $196(31) = 6104, \sqrt{6104} \approx 78.13$

Second term: $(361-196)(0) = 0$

So maximum value of $f(a)$ is about $78.13 + 0 = 78.13$?

Wait — this is less than 148!

But we need sum = 148

But the maximum value of each square root is about 78, so sum can't exceed about 156?

Wait: $\sqrt{a(a-165)}$, for $a = 196$: $196 \times (196-165) = 196 \times 31 = 6104, \sqrt{6104} \approx 78.13$

Similarly, maximum of other term at $a = 165$ is $\sqrt{196 \times 31} = \sqrt{6076} \approx 77.95$

So sum max is about $78 + 78 = 156$

So 148 is possible!

But at $a = 190$, we got only ~ 101 — why?

Wait: $a = 190, a(a-165) = 190 \times 25 = 4750, \sqrt{4750} \approx 68.9$

$(361-190) = 171, (196-190) = 6, 171 \times 6 = 1026, \sqrt{1026} \approx 32.0$

Sum ≈ 100.9

At $a = 180$: $51.96 + 53.82 = 105.78$

At $a = 170$: $a(a-165) = 170 \times 5 = 850, \sqrt{850} \approx 29.15$

$(361-170) = 191, (196-170) = 26, 191 \times 26 = 4966, \sqrt{4966} \approx 70.47$

Sum $\approx 29.15 + 70.47 = 99.62$

All are below 100!

Wait — this can't be. The maximum of the sum is about 156, but values are all below 100.

I must have made a calculation error.

Wait: when $a = 196$, first term: $a(a-165) = 196 \times (196-165) = 196 \times 31 = 6104, \sqrt{6104} = ?$

$6104: 78^2 = 6084, 79^2 = 6241 \rightarrow \sqrt{6104} \approx 78.13$

Second term: $(361-196)(196-196) = 165 \times 0 = 0 \rightarrow \text{sum} = 78.13$

At $a = 165$: first term = 0, second term: $(196)(31) = 6076, \text{sqrt} \approx 77.95 \rightarrow \text{sum} = 77.95$

So the maximum is about 78.13

But we need sum = 148

Impossible!

Contradiction.

So there is no solution?

But the problem states that such points exist.

Therefore, our assumption must be wrong.

Wait — is it possible that the point K is not the intersection of circle A and circle B , but another point?

The given is: $AK = AL = BK = CL = KL = 14$

We have:

$$- AK = 14 - BK = 14 - AL = 14 - CL = 14 - KL = 14$$

So yes, K is at distance 14 from both A and B , so lies on intersection

NudgeRL Solution

<start_working_out>

We are given a right triangle $\triangle ABC$ with $\angle A = 90^\circ$ and hypotenuse $BC = 38$. There are two points K and L inside the triangle such that:

$$AK = AL = BK = CL = KL = 14$$

We are to find the area of quadrilateral $BKLC$, which is expressed as $n\sqrt{3}$, and find n .

Step 1: Understand the configuration

We have: - $\angle A = 90^\circ$, so AB and AC are legs, $BC = 38$ is the hypotenuse. - Points K and L are inside the triangle. - All of the following distances are equal to 14: - $AK = AL = BK = CL = KL = 14$

This suggests symmetry. Since $AK = AL$, and $BK = CL$, and $KL = 14$, it's likely that K and L are symmetric with respect to the angle bisector or the median from A .

Moreover, $AK = AL$ suggests that K and L lie on the perpendicular bisector of BL or something symmetric.

But let's try to exploit symmetry.

Step 2: Consider symmetry

Since $AK = AL$ and $BK = CL$, and the triangle is right-angled at A , a natural symmetry is about the angle bisector of $\angle A$, which in a right triangle is the line $y = x$ if we place A at the origin and AB, AC along the axes.

Let's place the triangle in coordinate plane:

Let: - $A = (0, 0)$ - $B = (b, 0)$ - $C = (0, c)$

Then $BC = 38$, so by Pythagoras:

$$b^2 + c^2 = 38^2 = 1444$$

We are to find points K and L such that: - $AK = AL = 14$ - $BK = 14$ - $CL = 14$ - $KL = 14$

Let's suppose that due to symmetry, K and L are symmetric with respect to the line $y = x$. Then:

Let $K = (x, y)$, then $L = (y, x)$

Then: - $AK = \sqrt{x^2 + y^2} = 14 \Rightarrow x^2 + y^2 = 196$ — (1) - $BK = \sqrt{(x-b)^2 + y^2} = 14 \Rightarrow (x-b)^2 + y^2 = 196$ — (2) - $CL = \sqrt{(y-c)^2 + x^2} = 14 \Rightarrow (y-c)^2 + x^2 = 196$ — (3) - $KL = \sqrt{(x-y)^2 + (y-x)^2} = \sqrt{2}|x-y| = 14$

So:

$$\sqrt{2}|x-y| = 14 \Rightarrow |x-y| = \frac{14}{\sqrt{2}} = 7\sqrt{2}$$

So $x - y = \pm 7\sqrt{2}$. Without loss of generality, assume $x > y$, so $x - y = 7\sqrt{2}$ — (4)

Step 3: Use equations (1), (2), (3), (4)

From (1): $x^2 + y^2 = 196$

From (4): $x = y + 7\sqrt{2}$

Plug into (1):

$$\begin{aligned} (y + 7\sqrt{2})^2 + y^2 &= 196 \\ y^2 + 14\sqrt{2}y + 98 + y^2 &= 196 \\ 2y^2 + 14\sqrt{2}y + 98 &= 196 \\ 2y^2 + 14\sqrt{2}y &= 98 \\ y^2 + 7\sqrt{2}y &= 49 \end{aligned}$$

This is a quadratic in y . Let's solve it.

Complete the square:

$$\begin{aligned} y^2 + 7\sqrt{2}y &= 49 \\ y^2 + 7\sqrt{2}y + \left(\frac{7\sqrt{2}}{2}\right)^2 &= 49 + \left(\frac{7\sqrt{2}}{2}\right)^2 \\ y^2 + 7\sqrt{2}y + \frac{49 \cdot 2}{4} &= 49 + \frac{98}{4} = 49 + 24.5 = 73.5 \end{aligned}$$

So:

$$\left(y + \frac{7\sqrt{2}}{2}\right)^2 = 73.5 = \frac{147}{2}$$

So:

$$y = -\frac{7\sqrt{2}}{2} \pm \sqrt{\frac{147}{2}}$$

This is messy, and we haven't yet used the other conditions involving b and c . So perhaps we should go back to equations (2) and (3).

NudgeRL Solution (Cont.)

From equation (2): $(x - b)^2 + y^2 = 196$

From (1): $x^2 + y^2 = 196$

Subtract (1) - (2):

$$\begin{aligned}(x^2 + y^2) - [(x - b)^2 + y^2] &= 0 \\ x^2 - (x^2 - 2bx + b^2) &= 0 \\ x^2 - x^2 + 2bx - b^2 &= 0 \\ 2bx &= b^2\end{aligned}$$

So $b = 2x$

Similarly, go to equation (3): $(y - c)^2 + x^2 = 196$

From (1): $x^2 + y^2 = 196$

Subtract (3) - (1):

$$\begin{aligned}[(y - c)^2 + x^2] - [x^2 + y^2] &= 0 \\ (y - c)^2 - y^2 &= 0 \\ y^2 - 2cy + c^2 - y^2 &= 0 \\ -2cy + c^2 &= 0\end{aligned}$$

So $c = 2y$

Now we have: $-b = 2x - c = 2y - x^2 + y^2 = 196 - x - y = 7\sqrt{2}$

And also from the hypotenuse:

$$\begin{aligned}b^2 + c^2 &= 1444 \\ (2x)^2 + (2y)^2 &= 1444 \\ 4x^2 + 4y^2 &= 1444 \\ x^2 + y^2 &= 361\end{aligned}$$

Wait! This contradicts equation (1), which said $x^2 + y^2 = 196$.

So contradiction!

That means our assumption that $K = (x, y)$, $L = (y, x)$ is invalid?

But why? Because $AK = AL = 14$, and symmetry about $y = x$ makes sense, but the distance from K to B and L to C may not match unless the legs are in a certain ratio.

But we get a contradiction: $x^2 + y^2 = 196$ from $AK = 14$, but from $b^2 + c^2 = 1444$, we get $x^2 + y^2 = 361$, which is not equal.

So our assumption must be wrong.

Where did we go wrong?

Ah — in the step where we said from (2) and (1), we got $b = 2x$. Let's double-check:

From (1): $x^2 + y^2 = 196$

From (2): $(x - b)^2 + y^2 = 196$

Subtract (1) from (2):

$$\begin{aligned}(x - b)^2 - x^2 &= 0 \\ x^2 - 2bx + b^2 - x^2 &= 0 \\ -2bx + b^2 &= 0 \\ b(b - 2x) &= 0\end{aligned}$$

So either $b = 0$ (impossible) or $b = 2x$

Same for $c = 2y$

So the algebra seems correct.

But then $b^2 + c^2 = 4x^2 + 4y^2 = 4(x^2 + y^2) = 4 \times 196 = 784$

But we are told $BC = 38$, so $b^2 + c^2 = 38^2 = 1444$

But $784 \neq 1444$

Contradiction.

So our assumption that K and L are symmetric across $y = x$ must be invalid.

Alternative idea: Maybe K and L are symmetric with respect to the altitude from A , or the angle bisector?

Alternatively, perhaps the configuration is such that K and L are located so that they are equidistant from A , B , C , and each other.

Wait — we are given:

- $AK = AL = 14$ - $BK = 14$ - $CL = 14$ - $KL = 14$

So: - K is at distance 14 from A and B - L is at distance 14 from A and C - K and L are at distance 14 from each other

NudgeRL Solution (Cont.)

So K lies at intersection of circles: - Circle centered at A with radius 14 - Circle centered at B with radius 14
 Similarly, L lies at intersection of: - Circle centered at A with radius 14 - Circle centered at C with radius 14
 So both K and L are at intersection of circle A and circle B , and A and C respectively.
 So we can find coordinates.

Step 4: Coordinate geometry setup

Let's place: - $A = (0, 0)$ - $B = (b, 0)$ - $C = (0, c)$

Then $b^2 + c^2 = 38^2 = 1444$

Now, find intersection of circle centered at $A(0, 0)$ with radius 14 and circle centered at $B(b, 0)$ with radius 14.

Equation of circle at A : $x^2 + y^2 = 196$

Equation of circle at B : $(x - b)^2 + y^2 = 196$

Subtract:

$$\begin{aligned}(x - b)^2 + y^2 - x^2 - y^2 &= 0 \\ x^2 - 2bx + b^2 + y^2 - x^2 - y^2 &= 0 \\ -2bx + b^2 &= 0 \\ x &= \frac{b}{2}\end{aligned}$$

Plug into $x^2 + y^2 = 196$:

$$\begin{aligned}\left(\frac{b}{2}\right)^2 + y^2 &= 196 \\ \frac{b^2}{4} + y^2 &= 196 \\ y^2 &= 196 - \frac{b^2}{4}\end{aligned}$$

So $y = \pm \sqrt{196 - \frac{b^2}{4}}$

Since K is inside the triangle, we take $y > 0$, so

$$y_K = \sqrt{196 - \frac{b^2}{4}}$$

Similarly, find L : intersection of circle at A and circle at $C(0, c)$

Circle at C : $x^2 + (y - c)^2 = 196$

Subtract from $x^2 + y^2 = 196$:

$$\begin{aligned}x^2 + y^2 - [x^2 + (y - c)^2] &= 0 \\ y^2 - (y^2 - 2cy + c^2) &= 0 \\ 2cy - c^2 &= 0 \\ y &= \frac{c}{2}\end{aligned}$$

Then:

$$x^2 + \left(\frac{c}{2}\right)^2 = 196 \Rightarrow x^2 = 196 - \frac{c^2}{4}$$

So:

$$x_L = \sqrt{196 - \frac{c^2}{4}}$$

(positive, since inside triangle)

So: - $K = \left(\frac{b}{2}, \sqrt{196 - \frac{b^2}{4}}\right)$ - $L = \left(\sqrt{196 - \frac{c^2}{4}}, \frac{c}{2}\right)$

We are given that $KL = 14$

So compute distance between K and L :

$$KL^2 = \left(\frac{b}{2} - \sqrt{196 - \frac{c^2}{4}}\right)^2 + \left(\sqrt{196 - \frac{b^2}{4}} - \frac{c}{2}\right)^2 = 14^2 = 196$$

NudgeRL Solution (Cont.)

Let's define: $-p = \frac{b}{2} - q = \frac{c}{2}$

Then $b = 2p, c = 2q$

Then:

$$b^2 + c^2 = 4p^2 + 4q^2 = 4(p^2 + q^2) = 1444 \Rightarrow p^2 + q^2 = 361$$

Now express the coordinates: $-K = (p, \sqrt{196 - p^2}) - L = (\sqrt{196 - q^2}, q)$

Then:

$$KL^2 = (p - \sqrt{196 - q^2})^2 + (\sqrt{196 - p^2} - q)^2 = 196$$

Expand:

First term:

$$(p - \sqrt{196 - q^2})^2 = p^2 - 2p\sqrt{196 - q^2} + (196 - q^2)$$

Second term:

$$(\sqrt{196 - p^2} - q)^2 = (196 - p^2) - 2q\sqrt{196 - p^2} + q^2$$

Add both:

$$\begin{aligned} p^2 + 196 - q^2 + 196 - p^2 + q^2 \\ - 2p\sqrt{196 - q^2} - 2q\sqrt{196 - p^2} = 196 \end{aligned}$$

Simplify:

$$-p^2 - p^2 = 0 - -q^2 + q^2 = 0 - 196 + 196 = 392$$

So:

$$392 - 2p\sqrt{196 - q^2} - 2q\sqrt{196 - p^2} = 196$$

Bring to left:

$$\begin{aligned} 392 - 196 &= 2p\sqrt{196 - q^2} + 2q\sqrt{196 - p^2} \\ 196 &= 2p\sqrt{196 - q^2} + 2q\sqrt{196 - p^2} \\ 98 &= p\sqrt{196 - q^2} + q\sqrt{196 - p^2} \end{aligned}$$

Now recall that $p^2 + q^2 = 361$

Let's define $x = p, y = q$, so: $-x^2 + y^2 = 361 - x\sqrt{196 - y^2} + y\sqrt{196 - x^2} = 98$

Let's try to simplify or guess values.

Note that $x^2 + y^2 = 361$, so $x^2, y^2 \leq 361$

Also:

$$196 - x^2 \geq 0 \Rightarrow x^2 \leq 196$$

similarly $y^2 \leq 196$

Let's suppose $x = y$. Then:

$$2x^2 = 361 \Rightarrow x^2 = 180.5 \Rightarrow x \approx 13.43$$

Check if the equation holds:

Left side:

$$x\sqrt{196 - x^2} + x\sqrt{196 - x^2} = 2x\sqrt{196 - x^2}$$

Since:

$$x^2 = 180.5 \Rightarrow 196 - x^2 = 15.5$$

We have:

$$\sqrt{15.5} \approx 3.937$$

and

$$x \approx \sqrt{180.5} \approx 13.43$$

So:

$$2 \times 13.43 \times 3.937 \approx 2 \times 52.9 \approx 105.8 \neq 98$$

Too big.

So $x = y$ is too big.

We need smaller value.

Try to assume $x^2 = a, y^2 = 361 - a$

We want:

$$\sqrt{a}\sqrt{196 - (361 - a)} + \sqrt{361 - a}\sqrt{196 - a} = 98$$

Simplify:

$$196 - (361 - a) = a - 165$$

So:

$$\sqrt{a(a - 165)} + \sqrt{(361 - a)(196 - a)} = 98$$

NudgeRL Solution (Cont.)

Now define function:

$$f(a) = \sqrt{a(a-165)} + \sqrt{(361-a)(196-a)}$$

Domain:

$$a > 165$$

since:

$$a(a-165) \geq 0$$

Also:

$$a \leq 196, \quad 361 - a \geq 0 \Rightarrow a \leq 361$$

so:

$$a \in (165, 196]$$

Try $a = 169$:

$$-a(a-165) = 169(4) = 676, \quad \sqrt{676} = 26 - 361 - 169 = 192 - 196 - 169 = 27 - 192 \times 27 = 5184 - \sqrt{5184} = 72$$

Sum:

$$26 + 72 = 98$$

Perfect!

So:

$$a = 169$$

Hence:

$$x^2 = 169 \Rightarrow x = 13$$

and:

$$y^2 = 361 - 169 = 192 \Rightarrow y = \sqrt{192} = 8\sqrt{3}$$

Now check:

$$x^2 + y^2 = 169 + 192 = 361$$

Also:

$$\sqrt{a(a-165)} = \sqrt{169 \cdot 4} = \sqrt{676} = 26$$

and:

$$\sqrt{(361-169)(196-169)} = \sqrt{192 \cdot 27} = \sqrt{5184} = 72$$

Sum:

$$26 + 72 = 98$$

Perfect.

$$\text{So: } -p = x = 13 - q = y = \sqrt{192} = 8\sqrt{3}$$

$$\text{Then: } -b = 2p = 26 - c = 2q = 16\sqrt{3}$$

Check:

$$b^2 + c^2 = 26^2 + (16\sqrt{3})^2 = 676 + 256 \cdot 3 = 676 + 768 = 1444 = 38^2$$

Good.

Now we have coordinates:

$$K = \left(\frac{b}{2}, \sqrt{196 - \left(\frac{b}{2} \right)^2} \right) = (13, \sqrt{196 - 169}) = (13, \sqrt{27}) = (13, 3\sqrt{3})$$

and:

$$L = \left(\sqrt{196 - \left(\frac{c}{2} \right)^2}, \frac{c}{2} \right)$$

Since:

$$\frac{c}{2} = 8\sqrt{3}$$

and:

$$\left(\frac{c}{2} \right)^2 = (8\sqrt{3})^2 = 192$$

So:

$$\sqrt{196 - 192} = \sqrt{4} = 2$$

Thus:

$$L = (2, 8\sqrt{3})$$

NudgeRL Solution (Cont.)

Therefore: - $K = (13, 3\sqrt{3})$ - $L = (2, 8\sqrt{3})$

Now we want the area of quadrilateral $BKLC$

Points: - $B = (26, 0)$ - $K = (13, 3\sqrt{3})$ - $L = (2, 8\sqrt{3})$ - $C = (0, 16\sqrt{3})$

Quadrilateral $BKLC$ — in order:

$$B \rightarrow K \rightarrow L \rightarrow C \rightarrow B$$

We can compute its area using shoelace formula.

List the points in order:

1. $B = (26, 0)$ 2. $K = (13, 3\sqrt{3})$ 3. $L = (2, 8\sqrt{3})$ 4. $C = (0, 16\sqrt{3})$

Back to $B = (26, 0)$

Shoelace formula:

$$\text{Area} = \frac{1}{2} \left| \sum x_i y_{i+1} - \sum y_i x_{i+1} \right|$$

Compute:

$$\sum x_i y_{i+1}$$

$$- 26 \cdot 3\sqrt{3} = 78\sqrt{3} - 13 \cdot 8\sqrt{3} = 104\sqrt{3} - 2 \cdot 16\sqrt{3} = 32\sqrt{3} - 0 \cdot 0 = 0$$

Sum:

$$(78 + 104 + 32)\sqrt{3} = 214\sqrt{3}$$

Compute:

$$\sum y_i x_{i+1}$$

$$- 0 \cdot 13 = 0 - 3\sqrt{3} \cdot 2 = 6\sqrt{3} - 8\sqrt{3} \cdot 0 = 0 - 16\sqrt{3} \cdot 26 = 416\sqrt{3}$$

Sum:

$$(0 + 6 + 0 + 416)\sqrt{3} = 422\sqrt{3}$$

Now:

$$\text{Area} = \frac{1}{2} |214\sqrt{3} - 422\sqrt{3}| = \frac{1}{2} \cdot 208\sqrt{3} = 104\sqrt{3}$$

So:

$$n = 104$$

Final Answer:

<SOLUTION>104</SOLUTION>

G Full Evaluation Results

Table 3: p_{drop} ablation results. We report $\text{pass}@1$ estimated using 128 rollouts. Best results are represented as **bold**.

p_{drop}	AIME24	AIME25	AMC23	MATH500	Average
0.00	0.418	0.344	0.759	0.628	0.537
0.25	0.461	0.354	0.773	0.658	0.561
0.50	0.482	0.393	0.857	0.660	0.598
0.75	0.458	0.361	0.796	0.649	0.566

Table 4: Hint sampling ablation results. We report $\text{pass}@1$ estimated using 128 rollouts. Best results are represented as **bold**.

Sampling	AIME24	AIME25	AMC23	MATH500	Average
Random	0.482	0.393	0.857	0.660	0.598
Top ranked	0.448	0.355	0.774	0.632	0.552

H Broader Impacts

This paper proposes an efficient framework for structured exploration in reinforcement learning with verifiable rewards (RLVR). On the positive side, our method improves exploration efficiency without

Table 5: λ ablation results. We report $pass@1$ estimated using 128 rollouts. Best results are represented as **bold**.

λ	AIME24	AIME25	AMC23	MATH500	Average
0.9	0.403	0.366	0.806	0.648	0.556
1.0	0.436	0.359	0.831	0.643	0.567
1.1	0.482	0.393	0.857	0.660	0.598

Table 6: λ_{distill} ablation results. We report $pass@1$ estimated using 128 rollouts. Best results are represented as **bold**.

λ_{distill}	AIME24	AIME25	AMC23	MATH500	Average
0.0	0.423	0.362	0.826	0.628	0.560
0.1	0.482	0.393	0.857	0.660	0.598
0.5	0.425	0.361	0.730	0.629	0.536

relying on extremely large rollout budgets or expensive oracle supervision, which may help reduce the computational cost of training reasoning models and improve accessibility for smaller research groups.

However, improving exploration efficiency may also contribute to the development of increasingly capable reasoning systems, which could be misused in harmful or unintended ways. We therefore emphasize the importance of continued research on safety, oversight, and responsible deployment.

Table 7: ϵ_{high} scaling results. We report $\text{pass}@1$ estimated using 128 rollouts. Best results are represented as **bold**.

Algorithm	ϵ_{high}	AIME24	AIME25	AMC23	MATH500	Average
NudgeRL	0.2	0.482	0.393	0.857	0.660	0.598
GRPO	0.2	0.444	0.367	0.749	0.668	0.557
	0.24	0.451	0.373	0.795	0.645	0.566
	0.28	0.443	0.372	0.793	0.648	0.564
	0.32	0.452	0.358	0.813	0.640	0.566
	0.36	0.432	0.338	0.845	0.647	0.565
	0.40	0.406	0.341	0.781	0.638	0.541